

The Path of Carbon in Photosynthesis

VII. RESPIRATION AND PHOTOSYNTHESIS¹

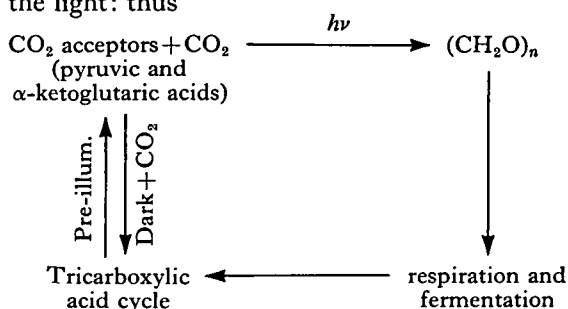
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PREVIOUS work (Calvin and Benson, 1948; Benson, Calvin, *et al.*, 1949) has shown that illumination of an algal suspension in the absence of carbon dioxide greatly enhances its ability to fix carbon dioxide in an immediately following dark period. The kinetics of the generation and decay of this ability have been determined, and it was shown that less than one minute of illumination was sufficient to bring this fixing ability almost to its saturation value.

Other experiments (Benson and Calvin, 1948) which determined the dependence of dark fixation rate on carbon dioxide pressure, showing it to be very similar to, if not identical with, the dependence of steady state photosynthesis upon carbon dioxide pressure, were used as a partial argument in support of the suggestion that the enhanced dark fixation immediately following illumination in the absence of carbon dioxide was indeed the process of carbon dioxide reduction taking place in photosynthesis.

Nevertheless, kinetic arguments involving unknown chemical reactions being what they are, it was still conceivably possible that this enhanced dark fixation was due only to the reversibility of the respiratory and fermentative decarboxylations. The photosynthesis occurring during pre-illumination presumably reduced the carbon dioxide partial pressure by reactions as yet unknown, thus shifting the fermentative and respiratory reaction equilibria in the direction of decarboxylation. Upon the introduction of carbon dioxide in the dark, the equilibria are shifted in the opposite direction, thus giving the enhanced fixation, but by processes presumed to be quite different from those taking place in the light: thus

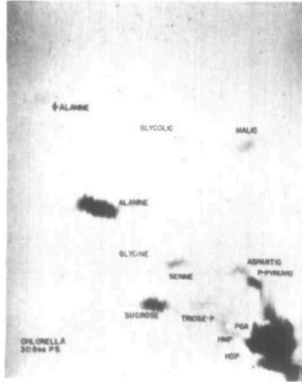


If this be the case, then the products formed in dark fixation of carbon dioxide should be the same following pre-illumination in the absence of carbon dioxide as they are following a dark saturation with carbon dioxide, and quite different from those formed in the light. If the pre-illuminated dark

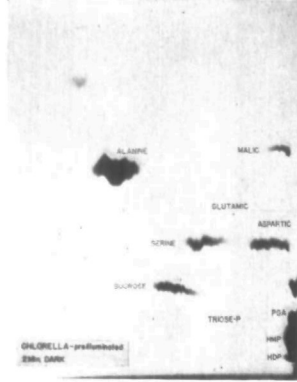
¹ This work was supported by the United States Atomic Energy Commission.

← Water-saturated phenol.

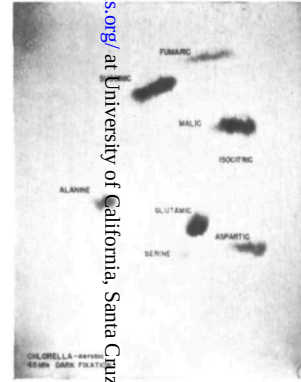
Butanol-water-propionic acid. →



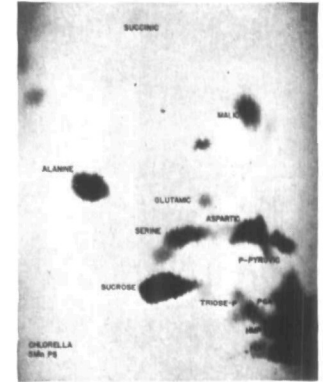
30 sec. PS. *Chlorella*.



2 min. dark pre-illuminated *Chlorella*.



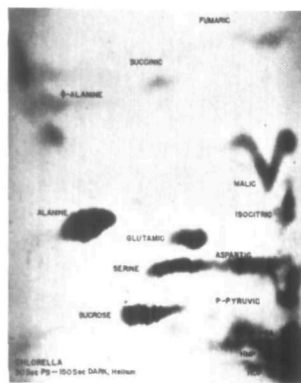
45 min. dark *Chlorella*.



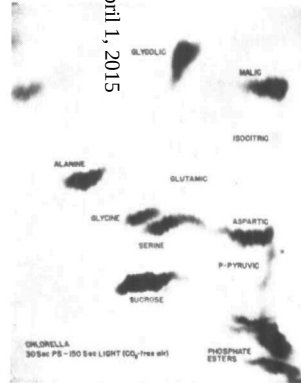
5 min. PS. *Chlorella*.



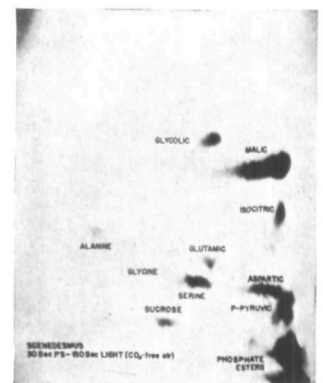
30 sec. PS.-150 sec. light (He) *Chlorella*.



30 sec. PS.-150 sec. dark (He) *Chlorella*.



30 sec. PS.-150 sec. light *Chlorella*.



30 sec. PS.-150 sec. light *Scenedesmus*.

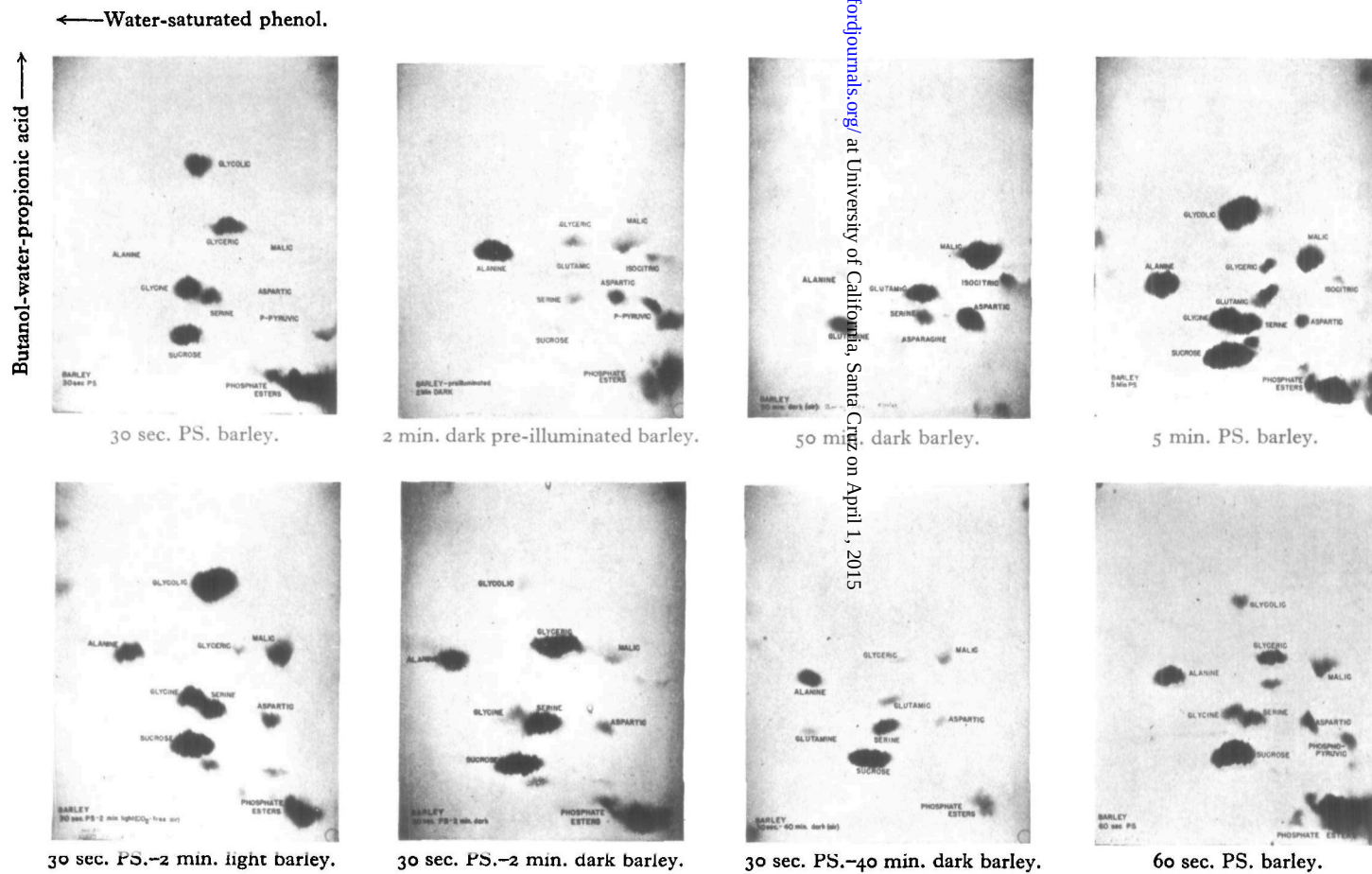


FIG. 1. C^{14} Radiograms of extracts of plants exposed to $C^{14}O_2$ under selected conditions. (For meaning of symbols, see Table I.)

fixation were the same process that takes place in ordinary photosynthesis, the products formed should be the same as those found in an equivalent period in the light and different from those formed in the dark without pre-illumination. If some of the compounds or closely related substances which appear in respiration and fermentation are also intermediates in photosynthesis, they should be formed in all three cases.

Figure 1 is a set of radiograms showing the products formed in each of the three cases for algae and for barley leaves (see 30 s. PS.C., 2 m. D.PI.C., 45 m. D.C., 30 s. PS.B., 2 m. D.PI.B., 50 m. D.B. in Table I). It is apparent that the products formed in dark fixation following pre-illumination in the absence of carbon dioxide correspond very closely to those formed in direct photosynthesis and not to all of those formed in dark fixation following dark saturation with carbon dioxide. These latter are indeed the substances expected from the simple reversibility of respiratory and fermentative reactions. We can thus confirm the suggestion, at first made on the basis of kinetic studies (Benson and Calvin, 1948), that all of the reactions lying between carbon dioxide and sucrose are dark reactions. The reducing energy required to achieve this transformation is supplied by the photochemical reaction involving the photolysis of water. Furthermore, the reducing power so provided is in the form of a definite chemical species rather than in the form of some excited electronic state of a molecule. This conclusion has been reached previously as a result of studies of fluorescence (Wassink and Katz, 1939) and of the comparative biochemistry of certain bacteria (Van Niel, 1941). The continued carbon dioxide absorption in the dark immediately following a strong illumination (McAlister and Myers, 1940; Emerson and Lewis, 1941) constitutes the direct observation of the effect. It is of interest to note that the successful separation of the photochemical apparatus for the splitting of water and the evolution of oxygen (Hill, 1937, 1939, Hill and Scarisbrick, 1940) from the carbon dioxide reducing system has been accomplished.

Figure 1 shows that a number of radioactive compounds are common to all three of the radiograms. These are, particularly, alanine, serine, aspartic acid, and malic acid. They may be taken as indicators for the presence of the corresponding keto-acids, pyruvic and oxalo-acetic acids, under all three circumstances. It is thus apparent that some compounds are involved as intermediates in both the photosynthetic and the respiratory cycles. Whether the common reservoirs of the two cycles are identical or are physically separate in the organism remains to be determined.

A number of experiments have been done in an attempt to determine something about the relationship between the photosynthetic and respiratory paths. A simple design for such an experiment would be to introduce the label into some of the photosynthetic intermediates by a short period of photosynthesis in radioactive carbon dioxide¹ and follow this by a suitable period of respiration in either dark or light. The result of such an experiment on algae is shown

¹ The experimental methods have been described previously: M. Calvin and A. A. Benson (1949); A. A. Benson *et al.*, *J. Amer. Chem. Soc.* (in press).

TABLE I

Percentage distribution of radioactivity among aqueous alcohol-soluble non-lipid products

	Alanine	Aspartic	Glutamic	Glutamine	Glycine	Serine	Succinic	Fumaric	Glycolic	Isocitric (citric?)	Malic	Sucrose	Phosphate esters
30 s. PS.C. ^a	6.1	0.8	—	—	—	1.3	—	—	0.5	—	0.9	0.5	89 ^f
2 m. D.PI.C. ^a	39	5.5	0.4	—	—	7.3	0.4	—	—	0.2	2.4	6.7	38
45 m. D.C. ^a	11	8.7	23	1.5	—	4.0	17	7.5	—	1.9	21	1.2	1.2
30 s. PS.B. ^b	1.1	0.1	—	—	12	4.9	—	—	8.3	—	0.5	15	59 ^m
2 m. D.PI.B. ^a	33	3.4	1.2	—	—	2.2	—	—	—	—	4.9	1.9	53 ^m
50 m. D.B. ^a	2.2	28 ^g	17	8.2	—	4.7	—	—	—	6	31	—	2.9
30 s. PS.-150 s. L.(He)C. ^a	23	17	0.4	—	—	3.8	—	—	—	—	8.4	27	21 ^p
30 s. PS.-150 s. D.(He)C. ^a	27	3.6	3.1	—	—	4.5	—	0.9	—	0.5	3.4	7.2	50 ^q
30 s. PS.-150 s. L.C. ^a	8.0	6.1	0.5	—	4.1	8.3	—	—	6.7	0.6	6.7	31	25 ^r
30 s. PS.-150 s. L.Sc. ^a	1.2	3.9	1.5	—	0.9	3.7	—	—	3.7	2.2	25	1.9	56 ^s
5 m. PS.C. ^a	1.9	3.0	0.4	—	—	2.0	0.2	—	—	0.1	1.9	23	65 ^t
30 s. PS.-2 m. L.B. ^b	2.2	1.4	—	—	7.4	6.7	—	—	23	—	2.3	45	11 ^u
30 s. PS.-2 m. D.B. ^b	4.6	0.8	—	—	0.7	17	—	—	0.2	0.5	0.7	34	42 ^v
30 s. PS.-40 m. D.B. ^b	6.4	1.6	2.3	1.4	—	7.3	—	—	—	0.8	2.1	62	16 ^w
60 s. PS.B. ^b	7.7	2.6	—	—	3.3	3.7	—	—	2.0	—	3.3	27	50 ^x
5 m. PS.B. ^b	8.1	0.7	0.8	—	4.5	4.3	—	—	6.4	0.2	2.6	65	6 ^y

Percentages were determined by direct counting of areas on the paper chromatogram defined by the radiogram.

s., seconds; m., minutes; PS., photosynthesis with C¹⁴O₂; C., *Chlorella pyrenoidosa*; Sc., *Scenedesmus D 3*; L., light; D., dark; B., barley seedling leaves; PI., 5–10 min. pre-illumination in helium; He, helium.

All treatments subsequent to photosynthesis were done in CO₂-free air unless specified as in helium.

^a In these experiments less than 5% of the total radioactivity fixed was insoluble in alcohol-water. ^b 2.5% insol. ^c 25% insol. ^d 6% insol. ^e 30% insol. ^f 30% insol. ^g 45% insol. ^h 10% insol. ⁱ 6% insol. ^j 3.2% insol. ^k 4.5% insol. ^l includes: Phosphoglyceric acid 24% (PGA), Hexose monophosphates 60% (HMP), Hexose diphosphates 3.9% (HDP), Triose-phosphates 0.5% (TrP). ^m includes free glyceric acid 10%. ⁿ includes free glyceric acid 2%. ^p includes PGA 1.3%, HMP 14%, HDP 4%, P. Pyruvic 1.8%. ^q includes PGA 6%, HMP 35%, HDP 7%, P. Pyruvic 1.8%. ^r includes PGA 3.2%, HMP 13%, HDP 7%. ^s includes PGA 25%, HMP 11%, HDP 18%, P. Pyruvic 2.2%. ^t includes PGA 26%, HMP and HDP 39%, P. Pyruvic 0.6%. ^u includes free glyceric acid 0.7%. ^v includes free glyceric acid 15%. ^w includes free glyceric acid 9%. ^x includes free glyceric acid 6%. ^y includes free glyceric acid 0.6%. ^z includes asparagine 2%.

in Fig. 1. One effect is immediately apparent. In those cases in which the algae have not been given a dark period either during or after exposure to C¹⁴O₂ there is little or no labelled glutamic and isocitric acid formed (see 30 s. PS.C., 30 s. PS.-150 s. L.(He)C., and 5 m. PS.C., Table I). If, however, they are given as little as 150 sec. of dark time following 30 sec. of photosynthesis in C¹⁴O₂, relatively large amounts of labelled glutamic and isocitric acids appear (see 30 s. PS.-150 s. D.(He)C., Table I). If the dark period is aerobic, the amounts of these compounds which appear are somewhat larger than under anaerobic conditions. It is thus clear that the light not only initiates a series of reactions constituting the photosynthetic reduction of carbon

dioxide but also in some way inhibits certain other reactions. Although the nature of the inhibition is as yet unknown, it is clear that light prevents some of the newly formed photosynthetic intermediates from participating in the tricarboxylic acid cycle as represented by glutamic and isocitric acids. In the barley leaves the formation of labelled glutamic acid is so slight even after relatively long dark periods that we cannot yet say whether the same phenomenon occurs or not. It is indicated by a comparison of the isocitric contents of 30 s. PS.B., 5 m. PS.B., 30 s. PS.-2 m. L.B., and 30 s. PS.-2 m. D.B., in which the last contains more of this acid than any of the first three. In the barley experiments another effect can be seen, which involves the conditions under which labelled glycine and glycolic acid are found. Glycine and glycolic acid, formed in the light from radiocarbon dioxide, disappear in as little as 2 min. in the dark. They appear in detectable amounts in as short a period as 5 sec. of photosynthesis in radiocarbon dioxide and are maintained in the light in the absence of carbon dioxide with the glycolic acid increasing (see 30 s. PS.B., 30 s. PS.-2 m. D.B., and 30 s.-2 m. L.B.). The same effect has been observed with algae. Additional experiments with algae showed that the presence of oxygen during the subsequent illumination enhanced the formation of labelled glycine and glycolic acid. In the case of glycine, for which the total amount present may be estimated from the amount of the ninhydrin colour produced on the paper, it appears that the short dark periods reduce not only the amount of labelled glycine but the total glycine as well. This would seem to indicate that the size of the reservoirs of free glycine and presumably of glycolic acid vary considerably with illumination and may possibly be related to the photosynthetic path of carbon. Degradation studies now under way should lead to a detailed exposition of the sequence of compounds involved.

LITERATURE CITED

- BENSON, A. A., and CALVIN, M. (1948). The path of carbon in photosynthesis. III. *Cold Spring Harbour symposia on quantitative biology*, **13**, 6.
- BENSON, A. A., CALVIN, M., *et al.* (1949). C^{14} in photosynthesis. Chapter 19, *Photosynthesis in plants*. Iowa State College Press.
- BENSON, A. A., *et al.* The path of carbon in photosynthesis. V. Paper chromatography and radioautography of the products. *J. Amer. Chem. Soc.* (in press).
- CALVIN, M., and BENSON, A. A. (1948). The path of carbon in photosynthesis. I. *Science*, **107**, 476.
- (1949). The path of carbon in photosynthesis. II. Intermediates in sucrose synthesis. *Science*, **109**, 140.
- EMERSON, R., and LEWIS, C. M. (1941). Carbon dioxide exchange and the measurement of quantum yield of photosynthesis. *Amer. J. Botany*, **28**, 789.
- HILL, R. (1937). Oxygen evolved by isolated chloroplasts. *Nature*, **139**, 881.
- (1939). Oxygen produced by isolated chloroplasts. *Proc. Roy. Soc. (Lond.)*, *B*, **127**, 192.
- and SCARISBRICK, R. (1940). Production of oxygen by illuminated chloroplasts. *Nature*, **146**, 61.
- MCALISTER, E. D., and MYERS, J. (1940). The time course of photosynthesis and fluorescence. *Smithsonian Inst. Publ. (Misc. Coll.)*, **99**, No. 6.
- VAN NIEL, C. B. (1941). The bacterial photosyntheses and their importance for the general problem of photosynthesis. *Adv. Enzymology*, **1**, 263.
- WASSINK, E. C., and KATZ, E. (1939). The initial changes in chlorophyll fluorescence in *Chlorella*. *Enzymologia*, **6**, 145.